KALDRIX DATA ANALYSIS MASTERY ROADMAP

10 weeks · 50 practical learning days · Beginner → Intermediate → Advanced → Pro → Expert

The programme develops real analytical capability through practical data, trainer review, QA and progressively stronger tools.

Week 1 - Beginner

Day	Lesson	Tool	Exercise
1	Spreadsheet Anatomy & Data Types	Google Sheets	Build and format a small participant register.
2	SUM, AVERAGE, MEDIAN, MIN, MAX & COUNT	Google Sheets	Summarize test scores and explain mean vs median.
3	Percentages & Relative References	Google Sheets	Calculate attendance and completion percentages.
4	IF Logic for Data Classification	Google Sheets	Create rule-based status fields.
5	Sort, Filter & Conditional Formatting	Google Sheets	Find top/bottom performers and filtered groups.

Week 2 - Beginner

Day	Lesson	Tool	Exercise
6	COUNTIF, SUMIF & AVERAGEIF	Google Sheets	Build a category summary table.
7	Text Cleaning: TRIM, PROPER & Standardization	Google Sheets	Clean a messy participant register.
8	Dates, Missing Values & Duplicates	Google Sheets	Audit and clean a registration dataset.
9	Charts & Evidence-Based Findings	Google Sheets	Create charts and evidence-based findings.
10	Beginner Capstone: One-Table Analysis	Google Sheets	Beginner capstone: training performance report.

Week 3 - Intermediate

Day	Lesson	Tool	Exercise
11	Absolute References, AND & OR	Google Sheets	Build a reusable staff eligibility model.
12	COUNTIFS, SUMIFS & AVERAGEIFS	Google Sheets	Create a multi-condition monitoring summary.
13	VLOOKUP & Exact Matching	Google Sheets	Enrich transaction data from a master list.
14	XLOOKUP & IFERROR	Google Sheets	Create resilient lookup logic.
15	Cross-Sheet Reconciliation	Google Sheets	Reconcile registration vs attendance.

Week 4 - Intermediate

Day	Lesson	Tool	Exercise
16	Pivot Tables	Google Sheets	Create operational Pivot Tables.
17	Pivot Charts, Slicers & Interactive Views	Google Sheets	Build an interactive operations view.
18	Data Quality Audit	Google Sheets	Audit a messy operational dataset.
19	KPI Design & Business Questions	Google Sheets	Create a KPI definition and calculation sheet.
20	Intermediate Capstone	Google Sheets	Intermediate capstone: multi-sheet monitoring system.

Week 5 - Advanced

Day	Lesson	Tool	Exercise
21	SQL SELECT, WHERE & ORDER BY	SQL	Query customer/program transactions.
22	SQL Aggregation & GROUP BY	SQL	Create grouped SQL summaries.
23	SQL JOINs	SQL	Build joined analytical datasets.
24	SQL CTEs & Window Functions	SQL	Create ranking and running-total queries.
25	Power Query: Import & Transform	Power BI / Power Query	Build a repeatable Power Query cleaning pipeline.

Week 6 - Advanced

Day	Lesson	Tool	Exercise
26	Power BI Data Modeling	Power BI	Create a simple star schema.
27	DAX Measures & Filter Context	Power BI / DAX	Create validated KPI measures.
28	Statistics for Analysts	Statistics	Perform an exploratory statistical analysis.
29	Dashboard Design & Data Storytelling	Power BI / Sheets	Build a decision-focused dashboard.
30	Advanced Capstone	SQL + Power BI	Advanced capstone: SQL-to-dashboard analysis.

Week 7 - Pro

Day	Lesson	Tool	Exercise
31	Python & pandas Foundations	Python / pandas	Reproduce spreadsheet analysis in pandas.
32	pandas Cleaning & Missing Data	Python / pandas	Build a reproducible pandas cleaning pipeline.
33	pandas Merge, GroupBy & Pivot	Python / pandas	Create an integrated pandas analysis.
34	Python Visualization	Python / matplotlib	Produce an analytical visualization pack.
35	Exploratory Data Analysis & Anomaly Detection	Python / Statistics	Perform structured EDA and anomaly investigation.

Week 8 - Pro

Day	Lesson	Tool	Exercise
36	Automation & Reproducible Reporting	Python	Automate a recurring analysis.
37	Experimentation & A/B Test Thinking	Statistics / Python	Evaluate a simple experiment.
38	Forecasting Fundamentals & Limits	Python / Analytics	Create and evaluate a baseline forecast.
39	Governance, Privacy, Versioning & Documentation	Data Governance	Create a governance and auditability pack.
40	Pro Capstone	Python + BI + QA	Pro capstone: automated analytical pipeline.

Week 9 - Expert

Day	Lesson	Tool	Exercise
41	Analytical Problem Framing	Analytical Thinking	Write an analysis specification.
42	Multi-Source Architecture & Lineage	Data Architecture	Create a data-flow and lineage design.
43	Metric Layer & Semantic Consistency	Analytics Engineering	Create a governed KPI/metric dictionary.
44	Causal Reasoning & Decision Risk	Statistics / Reasoning	Write a causal reasoning memo.
45	Executive Communication & Analytical Narrative	Communication	Produce an executive analytical brief.

Week 10 - Expert

Day	Lesson	Tool	Exercise
46	Model QA, Audit & Reconciliation	QA / Audit	Create a model QA and audit report.
47	AI-Assisted Analytics Workflow	AI + Analytics	Create an AI-assisted analysis verification log.
48	Live Change Request & Regression Testing	Integrated Analytics	Implement and validate a change request.
49	Stakeholder Defense & Technical Review	Communication / QA	Complete an oral defense rehearsal.
50	Final Expert Capstone	Integrated Data Analysis	Final expert capstone.